Fake Job Posting Detection

Milestone 3: Model Architecture & End-to-End Pipeline

The Deception

Fraudulent job postings manipulate missing metadata and utilize persuasive, urgent free-text to bypass traditional filters. The core challenge is extracting subtle, contextual semantic cues from highly heterogeneous and deeply imbalanced data.

The Architecture

An end-to-end NLP pipeline leveraging a fully fine-tuned RoBERTa-base classifier. Optimized with Focal Loss, the architecture maps diverse textual signals into a unified tensor space to aggressively isolate complex linguistic fraud patterns.

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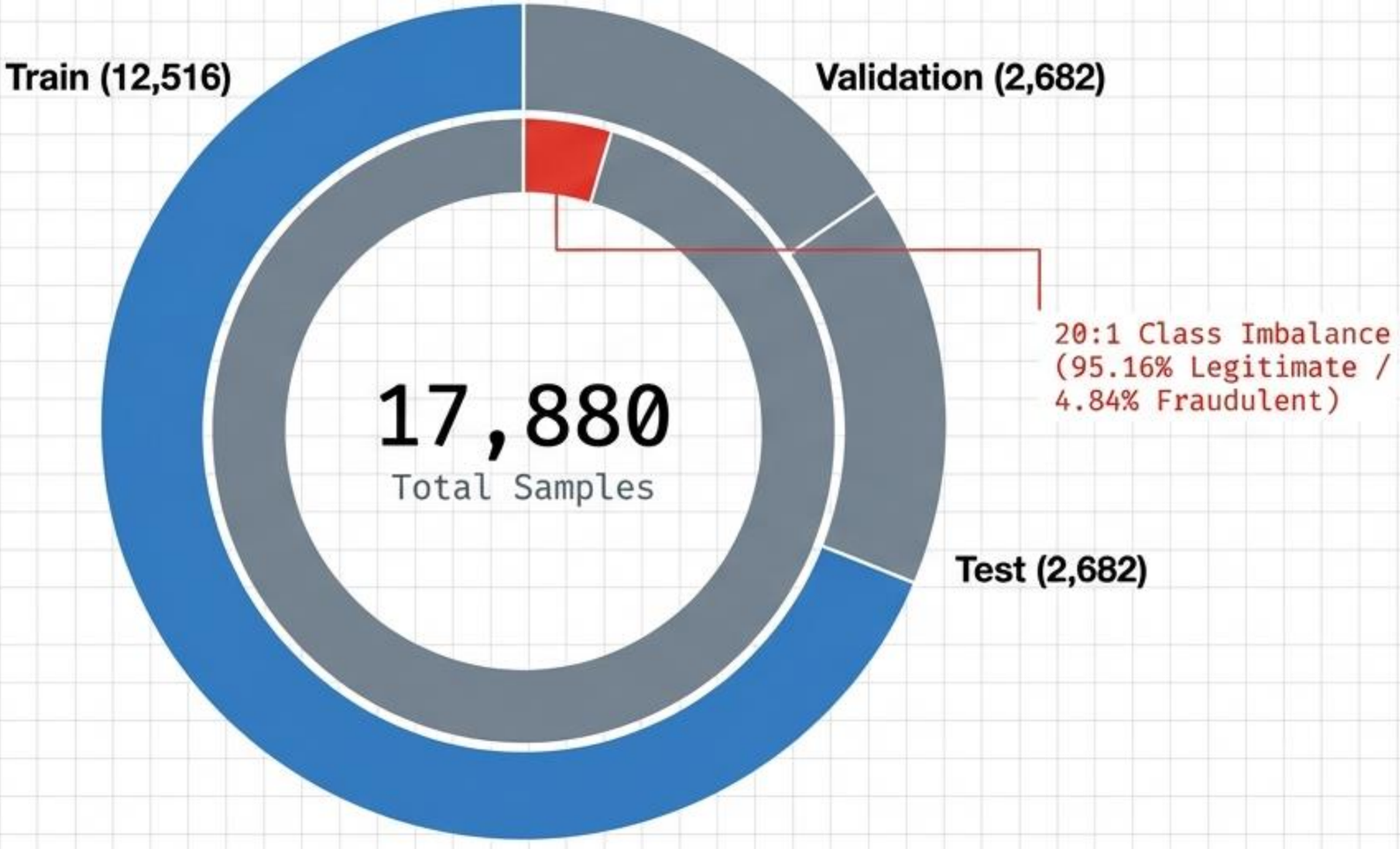
TECHNICAL SCHEMATIC SLIDE DIAGRAM

Preserving the Signal: Stratified Dataset



Deterministic stratified sampling guarantees the model learns and evaluates on real-world threat proportions.

- data/raw/
  (Unstructured CSVs)
- data/processed/
  (Stratified splits)
- models/outputs/
  (Tensors & Checkpoints)

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Dataset Organization & Rigorous Stratification

JetBrains Mono

Total Raw Dataset
17,880 Samples

Micro-typography

Training Set (70%)
12,516 samples (~606 fraud)

4.84% Fraud Rate Maintained

Micro-typography

Validation Set (15%)
2,682 samples (~130 fraud)

4.84% Fraud Rate Maintained

Micro-typography

Test Set (15%)
2,682 samples (~130 fraud)

4.84% Fraud Rate Maintained

Calltout box

> **Key Insight:** Stratified sampling guarantees the extreme 20:1 class imbalance is perfectly preserved across all splits, ensuring the model's evaluation metrics reflect true, real-world deployment conditions.

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Preprocessing Flow: Metamorphosis to Tensors

Node 1: Raw Input

Ingests unstructured text & missing metadata.

Insight: Missing values are NOT imputed. Absence is treated as a strong predictive signal.

Node 2: Formatting & Concat

Inter: Maps structured metadata to key-value pairs to preserve semantics. Joined using the [SEP] delimiter.

Rule: Metadata prioritized at the start of the sequence to prevent truncation.

Node 3: Tokenization

Inter: Applies RoBERTa BPE Tokenizer.

`max_length=512, truncation=True, padding='max_length'`

Node 4: Tensor Construction

Inter: Outputs mathematically unified structures:

`input_ids, attention_mask, labels`

Inter: This pipeline guarantees heterogeneous signals (salary, title, raw description) are converted into a unified contextual sequence expected by the transformer.

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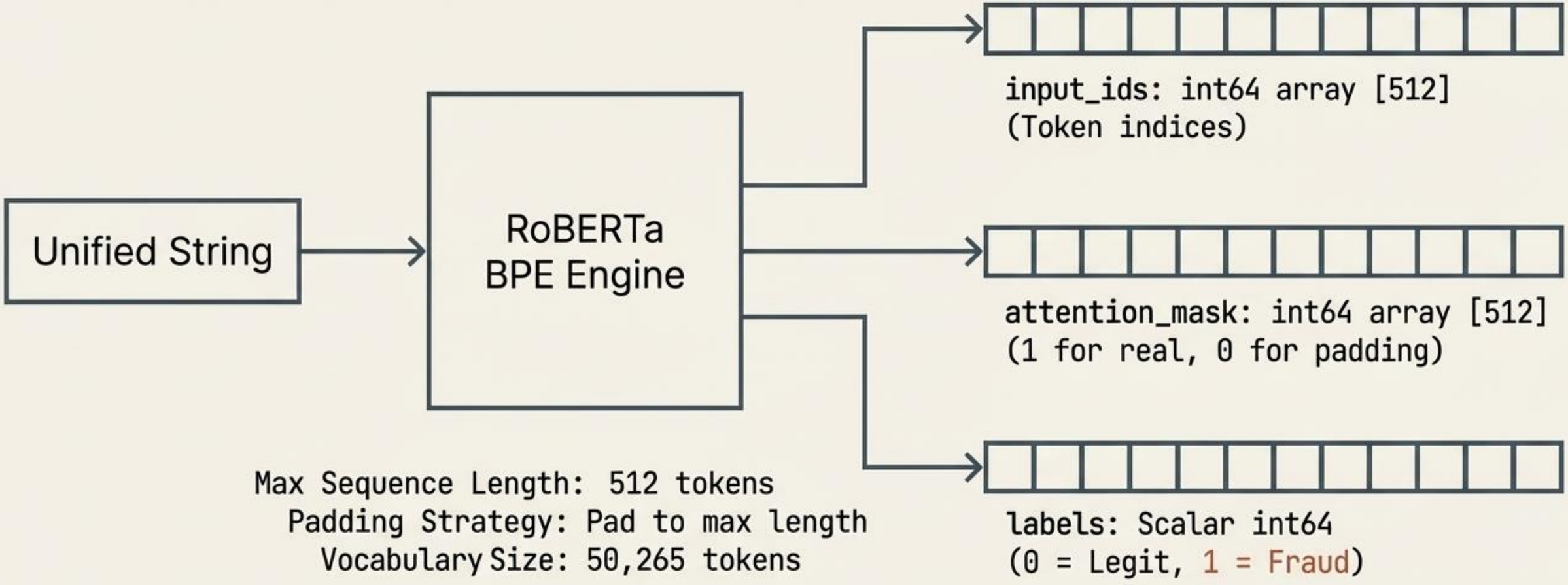
Tokenization & Tensor Construction



Max Sequence Length: 512 tokens
Padding Strategy: Pad to max length
Vocabulary Size: 50,265 tokens

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125M Parameter Architecture & Data Flow

Layer 6: Classification Head	Linear mapping from 768 dimensions → 2 logits (Fraud, Legit) (Params: ~1.5K).
Layer 5: Regularization	Dropout applied at p=0.1 to prevent overfitting.
Layer 4: [CLS] Pooling	Extracts sequence-level representation from the first token (Params: ~0.6M).
Layer 3: Transformer Encoder	12 stacked layers, 12 self-attention heads, 3072 intermediate feed-forward dims (Params: ~85.1M).
Layer 2: Token Embeddings	Translates indices to 768-dimensional dense vectors (Params: ~23.8M).
Layer 1: Input Tensors	Batched as [batch_size, 512]. Vocabulary size: 50,265.

100% Trainable Parameters.

The team **opted against frozen backbones**, fine-tuning all **125M parameters** to aggressively adapt representations to specific fraud patterns.

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Architecture Justification: Why RoBERTa?

Dimension	RoBERTa (Selected)	TF-IDF (Baseline)
Contextual Understanding	Full bidirectional context across 512 tokens. Understands sarcasm and urgency.	Bag-of-words. Ignores word order and context entirely.
Cross-field Reasoning	Correlates signals across title, description, and salary in a single self-attention pass.	Processes each feature independently.
Feature Engineering	Learns semantic features end-to-end automatically.	Requires exhaustive, manual hand-crafting of features.

Full Fine-Tuning vs. LoRA

Parameter-efficient methods like LoRA freeze the backbone. Because fraud data is highly domain-specific and severely imbalanced, Full Fine-Tuning was selected. It allows every layer to deeply adapt, significantly boosting recall on the hard minority class.

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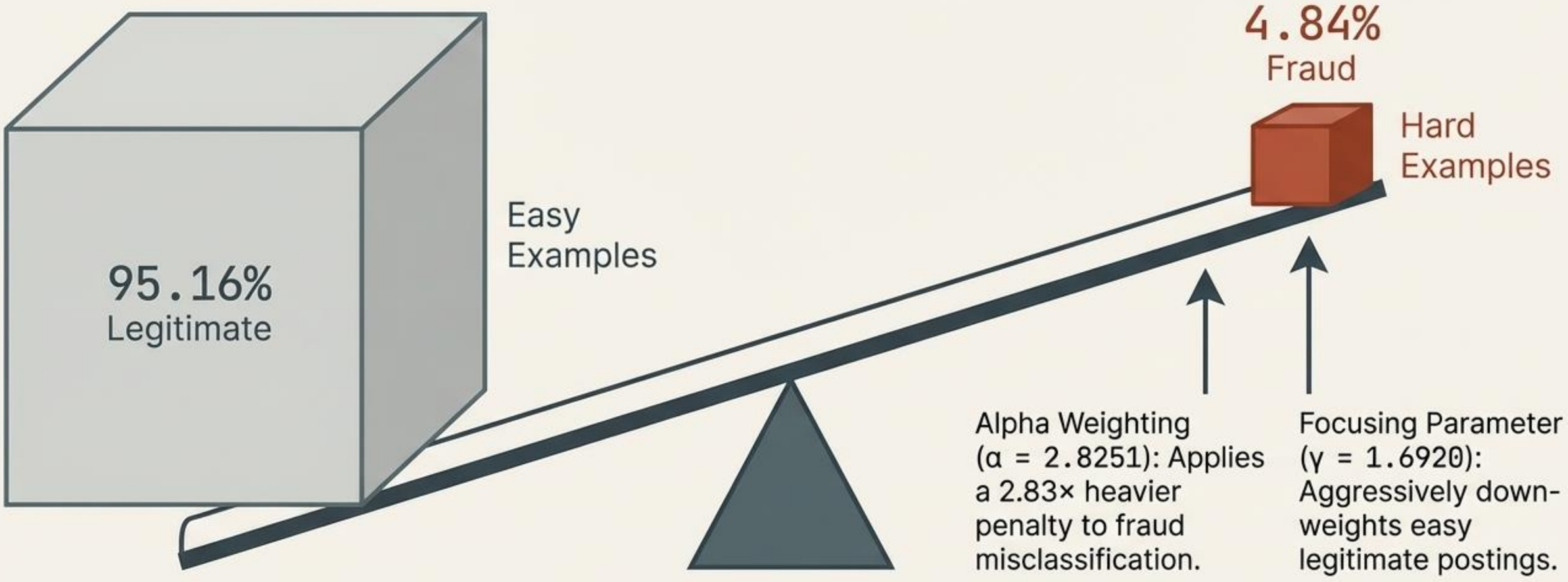
Overcoming the 20:1 Imbalance with Focal Loss

Standard Cross-Entropy Failure:
Blindly achieves 95% accuracy
by ignoring fraud entirely.



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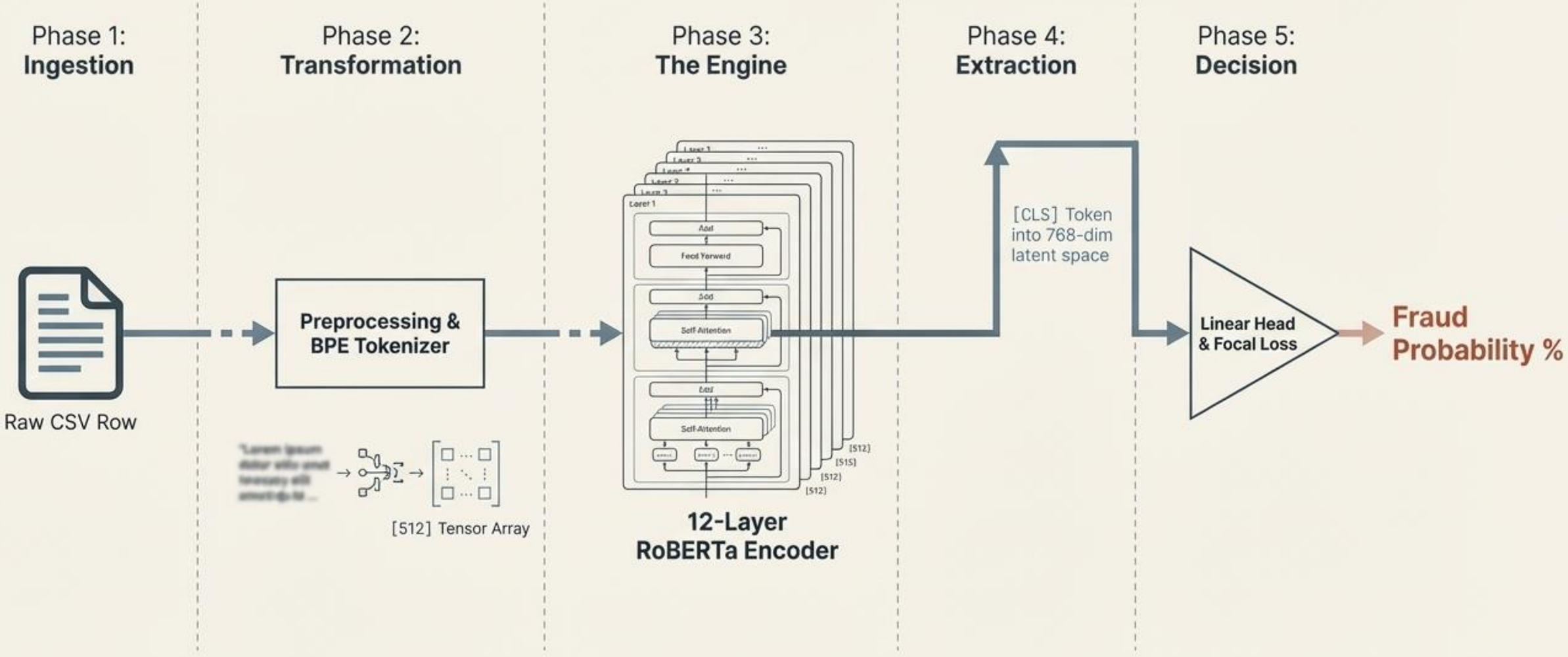
End-to-End Architecture Flow



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Unified Team Synergy: Milestone 3 Contribution Tracker

Core Collaborative Workstreams

End-to-End ML Pipeline

Jointly designed and verified the flow from raw data to model inference

RoBERTa Transformer Implementation

Collectively implemented and refined the model architecture for detection accuracy

Optimized Training Strategies

Combined efforts to validate focal loss and manage class imbalance

Arun Dutta & Hritik Roshan Maurya

Vivek Bajaj & Vishwas Mehta

deep learning solution for fake job detection

shared responsibilities' four member team

Joint Data Processing

Shared contributions in feature structuring, tokenization, and input consistency

Collaborative Quality Assurance

Every member contributed to debugging and pipeline verlfication for reproducibility

Shared Development & Quality

Combined efforts to validat to focal loss ands manage class for detediat accuracy

Peer-Driven Methodology

Achieved goals through regular code reviews, pair programming, and architectural discussions

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